THE Q-MODEL — From Q to Cosmology

A Reference Monograph on Structure, Complexity and Random Actualization

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This reference monograph accompanies the white paper *THE Q-MODEL — From Q to Cosmology* and is released for archival on Zenodo. It is *not* submitted for peer-reviewed journal publication. The central derivation claim (Einstein recovery from Q-events) remains *logical, not mathematical*: the metric and the BDG action are imported (proven but not TQM-derived), $G = \ell^2 c^3/\hbar$ is dimensional analysis, and no unique sharp prediction yet discriminates TQM from SM+ΛCDM. Every claim marked **verified** is backed by an executable xUnit test in `TQM.Tests/ResearchXC` (or `ResearchQG`); see Part V and Appendix A.

Abstract

This monograph is the reference edition of THE Q-MODEL (TQM), a theory of structure and content that compresses observable physics to four primitives — the individuation principle Q, Random Actualization, the scale triad $(\ell, \tau, \hbar)$, and a single continuous nonlinearity parameter M^2 — and holds that the *form* of every structure is derivable from these primitives while the *content* is realized by contingent draw. The monograph presents, in self-contained form: the formal primitives and dynamical system (the graph Laplacian L_Q and the Schrödinger equation); the verified continuum-limit chain from L_Q to the flat Laplacian, the d'Alembertian, the Laplace–Beltrami operator, the curved-space Schrödinger equation, and the Einstein tensor; the derivation hierarchy (gauge structure, spatial dimensionality, the multiplicity bound, the abundance law, phase-gradient gravity); the three-category taxonomy; the conditional no-go theorems; the quantitative predictions; and the complete inventory of executable verification results.

Contents

Part I

Foundations

Chapter 1

The Primitives

1.1 Statement of the theory

THE Q-MODEL rests on a single organizational claim, the **structure/content split**:

Structure is derivable; content is realized.

The *form* of observable structure (topology, symmetry, distribution shape, lower bounds) is provable from a minimal primitive set. The *content* (specific masses, couplings, multiplicities, angles) is a draw from a contingent ensemble and is therefore classified rather than derived.

1.2 The four primitives

Definition 1.1 (Primitive set)**.** *The theory admits exactly four primitives, and no others:*

1. *Q — the principle of individuation (ontology): the irreducible act by which a discrete event comes to be individuated from nothing.*

2. ***Random Actualization** — genuine ontological chance: given Q, the realized event locations are random within the causal structure. (Assumption **A-03**.)*

3. *$(\ell, \tau, \hbar)$ — the irreducible physical triple: a spacetime scale ℓ, a clock τ, and an action $\hbar$ (unit conventions).*

4. *M^2 — the single continuous nonlinearity parameter (the nonlinearity regime of the dynamics).*

No gauge-group, multiplicity, or hidden-dimension primitive is permitted. The primitives have two layers that meet at Q: an *ontology* layer (underwriting the derivation of structure) and a *dynamical* layer (underwriting the dynamical system of Chapter 3).

Remark 1.2. *Q individuates; Random Actualization realizes; $(\ell, \tau, \hbar)$ fixes units; M^2 sets the single continuous parameter that the derivation hierarchy pins to $M^2 \approx 5$ (Chapter 11).*

1.3 The quantum postulates (dynamical layer)

Axiom 1.3 (Q exists)**.** *Topological charge quanta have position x_i and phase θ_i, and interact pairwise with coupling $J_{ij} = f(|x_i - x_j|)$. (Assumed, irreducible.)*

Axiom 1.4 (Reversible dynamics)**.** *The state norm $\|\psi\|^2$ is conserved; equivalently, evolution is unitary.*

Axiom 1.5 (Born rule)**.** $P = |\langle\phi|\psi\rangle|^2$*, uniquely selected by additivity (Gleason's theorem, 1957).*

Axiom 1.6 (Measurement)**.** *A measurement yields one definite outcome (the collapse axiom).*

Postulates 1–2 derive the Hilbert space and the Schrödinger equation (Chapter 3); postulates 3–4 close the interpretation.

Chapter 2

Formalization of Q and Random Actualization

This chapter records the formal status of the primitives, as established by the formalization audits (executable; see Part V).

2.1 Status of Q

Component	Status	Content
Object	**Formalized**	a discrete event (individuated entity)
State space	**Formalized**	positions x_i and phases θ_i
Operations	**Formalized**	individuation, coupling J_{ij}
Configuration space	**Partial**	graph of events, no continuum limit native
Dynamics	**Partial**	L_Q generator (postulate 1)
Symmetries	**Partial**	S^1 phase, Z_2, permutation S_n
Continuum limit	**Partial**	verified for L_Q (Part II)
Measure	**Missing**	no measure space from primitives alone

Classification: Q is *partially formalized* — object, state space, and operations exist; the measure and the action functional remain as postulates.

2.2 Status of Random Actualization

Random Actualization is **A-03**, an assumption, not a derived object. The formalization audit found:

Component	Status	Content
Random variable	**Formalized**	the draw of event locations
Generator	**Formalized**	ontological chance
$(\Omega, \mathcal{F}, P)$	**Partial**	ensemble unspecified
Ensemble measure	**Partial**	no explicit measure

Classification: the random variable and generator are formalized; the probability space $(\Omega, \mathcal{F}, P)$ and the ensemble measure are only partially formalized.

Remark 2.1. *The hostile reviewer's charge that "Random Actualization is verbalism" is therefore* partially valid*: it is an* admitted *primitive (A-03), not a derived object, and the paper states this explicitly.*

Chapter 3

The Dynamical System

3.1 The graph Laplacian L_Q

Definition 3.1 (Graph Laplacian)**.** *For the Q-interaction graph with adjacency A and degree matrix D,*

$$L_Q = D - A. \tag{3.1}$$

L_Q is real, symmetric, positive semi-definite, and has zero row sums. Its eigenvectors form an orthonormal basis — the ***Hilbert space*** *of the theory.*

3.2 The tight-binding identity (TQM-142)

Proposition 3.2 (Tight-binding identity)**.** *On a 1D chain,*

$$(L_Q)_{ij} = 2\delta_{ij} - \delta_{i,j+1} - \delta_{i,j-1}, \tag{3.2}$$

which is identical *to the tight-binding Hamiltonian with on-site energy $\varepsilon = 2t$: $H = tL_Q$. This is a mathematical* ***identity****, not an analogy.*

3.3 Schrödinger from reversibility (TQM-149–151)

Theorem 3.3 (Schrödinger equation from reversibility)**.** *Consider linear evolution $\partial_t\psi = M\psi$ with conserved norm $\partial_t\|\psi\|^2 = \psi^\dagger(M + M^\dagger)\psi = 0$. Then*

$$M^\dagger = -M. \tag{3.3}$$

For real M, this is $M^T = -M$. The simplest 2×2 antisymmetric matrix J satisfies $J^2 = -I$ (so J acts as the imaginary unit), and

$$M = J \otimes L_Q \quad \Longrightarrow \quad i\partial_t\psi = L_Q\,\psi, \tag{3.4}$$

with unitary evolution

$$\psi(t) = e^{-iL_Q t}\,\psi(0). \tag{3.5}$$

3.4 Observables from the spectrum (TQM-145)

Observable	Definition
effective mass	$m_{\text{eff}} = 1/\lambda_1$
energy	$E = \text{tr}(L)$
gap	$\Delta = \lambda_2 - \lambda_1$
correlation length	$\xi = 1/\sqrt{\lambda_1}$
diffusion constant	$D = \lambda_1$
channel capacity	$C = \log_2 N$

Remark 3.4. *These are* dimensional assignments*, not dynamical predictions; the hostile reviewer correctly notes this is a scope boundary.*

3.5 The ontology → physics bridge

The bridge has two legs, both now tested:

1. L_Q eigenvector basis = Hilbert space (TQM-149); and
2. the causal-set continuum limit yields the metric (QG-001, XC006–012).

Part II

The Continuum-Limit Program (Verified)

Chapter 4

$L_Q \to -\nabla^2$: The Flat Laplacian

Theorem 4.1 (Discrete spectrum of the chain Laplacian)**.** *The graph Laplacian of a 1D chain of N sites has the exact spectrum*

$$\lambda_k = \frac{1}{\Delta x^2}\left[2 - 2\cos\frac{\pi k}{N+1}\right], \qquad k = 1, \ldots, N, \tag{4.1}$$

with Δx the lattice spacing. In the continuum limit $N \to \infty$, $\Delta x \to 0$ with the length fixed,

$$\lambda_k \to (\pi k)^2. \tag{4.2}$$

Proof. The chain Laplacian is the tridiagonal Toeplitz matrix with diagonal $2/\Delta x^2$ and off-diagonal $-1/\Delta x^2$; its eigenvectors are the discrete sine modes $\sin(\pi kj/(N+1))$. □

Verification. `GraphLaplacianContinuumTests` builds chains with $N = 32, 64, 128, 256$, diagonalizes L_Q via MathNet.Numerics, and compares against the closed form. The relative error *decreases* with N (second-order convergence, error $\sim 4\times$ per doubling), establishing that L_Q converges to the flat Laplacian with controlled error. **Result: PASSED.**

Chapter 5

BDG $\to \Box$: The d'Alembertian

The causal-set d'Alembertian is the Benincasa–Dowker–Glaser (BDG) operator: a directed, alternating sum over causal intervals of the sprinkling.

Theorem 5.1 (BDG convergence)**.** *The BDG operator converges to the continuum d'Alembertian $\Box$ as the sprinkling density grows, with leading truncation error $O(h^2)$.*

Verification. `BDGOperatorContinuumTests` verifies convergence with a tolerance 10^{-2} (the leading $O(h^2)$ error at $h = 1/16$ is $\approx 4.37 \times 10^{-3}$); the error falls $\sim 4\times$ per h-halving. **Result: PASSED.**

Chapter 6

The Quantum-Gravity Bridge

The two continuum chains are *partially connected* — and their signatures are incompatible.

Proposition 6.1 (Two distinct operators)**.** *1. L_Q is an* undirected, positive *graph Laplacian whose continuum limit is $-\nabla^2$ (Riemannian, elliptic, positive semi-definite).*

2. The BDG operator is a directed, alternating *causal-set operator whose continuum limit is $\Box$ (Lorentzian, hyperbolic, indefinite).*

Verification. `QuantumGravityBridgeTests` (three tests) establish:

1. L_Q is positive semi-definite (minimum eigenvalue ≥ 0);
2. the BDG operator is indefinite (possesses negative eigenvalues);
3. the two are incompatible in signature — the graph Laplacian cannot serve as a Lorentzian d'Alembertian (BdgUniquenessAnalyzer O3/O6 explicitly rejects it).

Result: PASSED (3/3).

Remark 6.2. *This is the single most important structural fact of the continuum program: TQM possesses* two *chain types, one Riemannian ($L_Q \to -\nabla^2$) and one Lorentzian (*BDG $\to \Box$*), which are* disjoint *in signature. The bridge between them is not yet closed natively.*

Chapter 7

The Weighted Laplacian and Laplace–Beltrami

7.1 The weighted Laplacian

Definition 7.1 (Weighted Laplacian). *Given the spatial coupling matrix K_{ij}, the weighted graph Laplacian is*

$$L_W = D_K - K, \qquad (D_K)_{ii} = \sum_j K_{ij}. \tag{7.1}$$

This is the standard unnormalized *graph Laplacian (Belkin–Niyogi), valid for uniform density.*

Proposition 7.2 (Properties). *L_W is symmetric, has zero row sum, is positive semi-definite, and reduces to the unweighted Laplacian when $K = A$.*

Verification. `WeightedLaplacianTests` (four tests) confirm symmetry, zero row sum, positive semi-definiteness, and reduction to the unweighted case. **Result: PASSED (4/4).**

7.2 The Laplace–Beltrami approximation

Theorem 7.3 (Laplace–Beltrami from L_W). *L_W approximates the Laplace–Beltrami operator Δ_g in the flat limit and reproduces standard manifold examples.*

Verification. `LaplaceBeltramiTests` (three tests):

1. *Flat limit*: L_W preserves the flat spectrum;
2. *Variable weights*: non-uniform K changes the spectrum;
3. *Known manifold*: the S^1 cycle graph reproduces the circle spectrum.

A subtle degeneracy was found and fixed by audit: the nonzero S^1 modes are two-fold degenerate ($e^{\pm ik\theta}$), so in ascending order the first occurrence of mode k is at index $2k-1$, not k. **Result: PASSED (3/3).**

Chapter 8

The Curved-Space Schrödinger Equation

Theorem 8.1 (Curved Schrödinger equation)**.** *The operator L_W defines a curved-space Schrödinger equation*

$$i\,\partial_t\psi = L_W\psi. \tag{8.1}$$

Because L_W is self-adjoint, the evolution is unitary and the norm is conserved:

$$\partial_t\|\psi\|^2 = 0. \tag{8.2}$$

Verification. `CurvedSchrodingerTests` (three tests): unitarity (norm conservation), reduction to the flat Schrödinger equation, and the existence of a curved operator. **Result: PASSED (3/3).**

Remark 8.2. *This establishes the* Schrödinger *side of the curved-space bridge: the weighted Laplacian L_W produces a well-defined, unitary curved Schrödinger equation. The* metric *that would make L_W a genuine Δ_g remains external (Chapter 10).*

Chapter 9

The Einstein Tensor Chain

9.1 Standard differential geometry

Definition 9.1 (Christoffel symbols)**.**

$$\Gamma^{\lambda}_{\mu\nu} = \tfrac{1}{2}\, g^{\lambda\sigma}\left(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}\right). \tag{9.1}$$

Definition 9.2 (Riemann curvature)**.**

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu \Gamma^{\rho}_{\nu\sigma} - \partial_\nu \Gamma^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma}. \tag{9.2}$$

Definition 9.3 (Ricci and Einstein tensors)**.**

$$R_{\sigma\nu} = R^{\rho}{}_{\sigma\rho\nu}, \qquad G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2} R\, g_{\mu\nu}. \tag{9.3}$$

9.2 Verified examples

Unit 2-sphere $g = \mathrm{diag}(1, \sin^2\theta)$ at $\theta = \pi/4$:

$$\Gamma^{\theta}_{\phi\phi} = -\sin\theta\cos\theta = -0.5, \qquad K = 1, \qquad R = 2K = 2, \qquad R_{\mu\nu} = K g_{\mu\nu}. \tag{9.4}$$

Unit 3-sphere $g = \mathrm{diag}(1, \sin^2\chi, \sin^2\chi \sin^2\theta)$ at $\chi = \theta = \pi/4$: $R_{\mu\nu} = 2g_{\mu\nu}$, $R = 6$, hence

$$G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2} R g_{\mu\nu} = -g_{\mu\nu} = -\mathrm{diag}(1, \tfrac{1}{2}, \tfrac{1}{4}). \tag{9.5}$$

Theorem 9.4 (2D Einstein tensor vanishes)**.** *In two dimensions $G_{\mu\nu} \equiv 0$ identically ($R_{\mu\nu} = \frac{1}{2} R g_{\mu\nu}$); a non-trivial $G_{\mu\nu}$ requires dimension ≥ 3.*

Verification. `EinsteinTensorTests` (four tests: metric→Christoffel, Christoffel→Riemann, Riemann→Ricci, Ricci→Einstein) and `EinsteinTensorIntegrationTests` (four tests on the integrated builder `EinsteinTensorBuilder`). **Result: PASSED (8/8).**

Remark 9.5 (Where the chain breaks in TQM)**.** *The standard chain is fully functional. But TQM's own analyzers (`QuantumGravityEmergenceAnalyzer`, `GrBridgeAnalyzer`)* describe *the chain as strings (`GeoStep`) and* never compute *it. The chain breaks at Step 1 —* metric → Christoffels *— because TQM possesses no native metric field $g_{\mu\nu}(x)$: it is imported (Chapter 10).*

Chapter 10

Metric Emergence and Origin Closure

10.1 The causal distance structure

Definition 10.1 (Causal volume and distance). *For a causal interval of depth D in d dimensions, the causal volume is*

$$N(D) \propto D^d, \tag{10.1}$$

and the causal distance is

$$L = N^{1/d}. \tag{10.2}$$

Verification. `MetricGenerationTests` and `MetricEmergenceTests` confirm: $L = N^{1/d}$ recovers the depth exactly ($d = 2, 3, 4$; $D = 1, \ldots, 8$); the dimension is recovered from volume growth; and the resulting distance matrix satisfies all four metric axioms (non-negativity, identity of indiscernibles, symmetry, triangle inequality). **Result: PASSED.**

10.2 The conformal factor from the counting measure

Theorem 10.2 (Conformal factor). *The counting measure fixes the volume element $\sqrt{|g|} = \rho$ (density ρ). For a conformally flat ansatz $g = f\,\eta$,*

$$\sqrt{|g|} = f^{\,d/2} = \rho \quad \Longrightarrow \quad f = \rho^{\,2/d}. \tag{10.3}$$

Verification. `MetricEmergenceTests` confirms the round-trip $f^{d/2} = \rho$ for $\rho \in \{0.5, 1, 2, 8\}$, $d \in \{2, 3, 4\}$. The conformally flat candidate $g = f\eta$ is constructible: constant f gives $R = 0$; $f = 1 + \frac{1}{2}x^2$ gives $R = -0.6145$, matching the Liouville result $K = -\frac{1}{2f}\nabla^2 \ln f$. **Result: PASSED.**

10.3 The conformal class: Malament's theorem

Theorem 10.3 (Malament 1977; Hawking–King–McCarthy 1976). *A bijection preserving the causal order $\prec$ between two past-and-future-distinguishing spacetimes is a conformal isometry: the causal order determines the metric* up to a conformal factor.

Verification. `ConformalStructureTests` and `MetricOriginTests` establish: the causal order is a valid partial order whose boundary is the light cone; the null structure is invariant under $g \to fg$ $(f > 0)$ but *changed* by a non-conformal rescale $g = \mathrm{diag}(-1, 2)$; hence the light cone picks out *exactly* the conformal class. **Result: PASSED.**

Theorem 10.4 (Metric origin closure)**.** *The conformal-class "gap" is an* imported theorem *(Malament), not a theory gap. The metric origin is closed:*

$$Q\text{-}events \to causal\ order\ (native) \to conformal\ class\ (proven) \to conformal\ factor\ (native) \to g_{\mu\nu}. \tag{10.4}$$

Classification of the metric origin:

Link	Status	Nature
Q-events	**NATIVE**	TQM-derived primitive
Causal order	**NATIVE**	precedence from individuation
Conformal class	**IMPORTED**	Malament/HKM — *proven*
Conformal factor	**NATIVE**	counting measure, $f = \rho^{2/d}$
Full $g_{\mu\nu}$	**determined**	class × factor

Remark 10.5. *The metric origin is* already solved*: importing a proven theorem is standard, not a defect. The genuinely open item is the* native *metric → operator coupling (the "G4" gap): TQM does not produce $g_{\mu\nu}$ from Q-events, it imports it.*

Part III

The Derivation Hierarchy

Chapter 11

Complexity and Spatial Dimensionality

11.1 The complexity functional

The complexity argument is a **weighted six-component decomposition** of the conditions for observers (`ComplexityOptimumAnalyzer.cs`), not a single scalar variational functional.

Component	Weight	M^2 window
Structure formation	3.0	2–8
Particle stability	2.5	3–7
Chemistry potential	4.0	3–5
Information capacity	2.0	3–10
Evolution potential	1.5	2–6
Observer viability	5.0	4–6

11.2 Spatial dimensionality $d = 3 + 1$

Theorem 11.1 (Unique dimensionality). $d = 3+1$ *is the* unique *value satisfying simultaneously: Bertrand's theorem (closed orbits only for* $1/r^2$*), Gauss's law (*$F \propto 1/r^{d-1}$*, Coulomb only in 3D), knot stability (codimension-2), Huygens (sharp propagation in odd d), and 2 GR polarizations.* $d = 2 + 1$ *fails gravity/atoms/knots;* $d \geq 4 + 1$ *fails orbits and knots.*

Remark 11.2. $M^2 \approx 5$ *is the observer-viability* intersection *of the component windows, not a hand-inserted number. Confidence T-02 = 0.85; this is a window-intersection argument, not a variational theorem. The analyzer records that the generation count* $G \approx 3$ *is a* plateau, *not a peak.*

Chapter 12

Gauge Structure

12.1 $U(1)$ — derived

Theorem 12.1 ($U(1)$ from the circle)**.** *Phase lives on S^1. Its isometry group is*

$$\mathrm{Aut}(S^1) = U(1), \qquad \pi_1(S^1) = \mathbb{Z}, \tag{12.1}$$

and the winding $\oint \nabla\theta \cdot dl = 2\pi n$ yields integer topological charge. (Confidence 0.95.)

12.2 $SU(2)$ and $SU(3)$ — real-underived structure

Factor	Status	Confidence
$U(1)$	**DERIVED**	0.95
$SU(2)$	**REAL-UNDERIVED** (emergent)	0.70
$SU(3)$ structure	**REAL-UNDERIVED** (emergent)	0.10
color count $n = 3$	**DRAWN**	—

Binary winding $\{n = \pm 1\}$ gives the minimal doublet; the Bloch sphere gives $SO(3) = SU(2)/Z_2$; the spinor double cover gives $SU(2)$. The automorphism of the n-defect moduli space satisfies $\mathrm{Aut}(C^n/S_n) \supseteq SU(n)$, deriving $SU(3)$ *structure* but not the count.

Remark 12.2. *TQM derives the gauge-group* ***structure*** *(which group),* ***not*** *the gauge* ***dynamics*** *(Maxwell / Yang–Mills actions are borrowed; the 8-gluon algebra is assumption A-07). The count $n = 3$ is unfixed (gauge-count no-go T-09, provisional 0.10).*

Chapter 13

Flavor and the Koide Relation

13.1 The Yukawa operator

The Yukawa operator is the overlap operator

$$Y_{ij} = \langle \mathrm{arch}_i \,|\, \mathrm{amplitude} \,|\, \mathrm{arch}_j \rangle. \tag{13.1}$$

Its *form* is derived; its *spectrum* (the hierarchy 1:207:3478) is underived — set by the contingent architecture shapes.

13.2 The Koide relation

Theorem 13.1 (Koide relation)**.** *The charged-lepton pole masses satisfy*

$$Q = \frac{\sum m}{(\sum \sqrt{m})^2} = 0.6666605, \qquad \theta = 44.9997^\circ, \tag{13.2}$$

equivalently $Q = 2/3$ at $\theta = \arccos(1/\sqrt{2}) = 45^\circ$. The relation is real ($\sim 10^{-5}$ precision; Bayes factor vs. coincidence $\approx 3.2 \times 10^4$), predictive (1981 prediction $m_\tau = 1776.97$ MeV, confirmed 1992), and RG-stable.

No-Go Theorem 13.2 (Koide origin — T-08)**.** *No symmetry (S_3), attractor, topology (S^1), or information-geometry selects $Q = 2/3$. (Confidence 0.70.)*

The closure audit (Phase 159) exhausted seven routes:

Route	Status
Symmetry (S_3)	No-Go
Topology (S^1)	No-Go
Attractors	No-Go
Information Geometry	No-Go
Group Theory	No-Go
$m = 3$ Closure	No-Go
Moduli Arguments	Blocked

The Koide origin is a *closed question* — real, predictive, RG-stable, *underivable* — the canonical **REAL-UNDERIVED** structure.

No-Go Theorem 13.3 (Neutrino-Koide — T-11, falsification)**.** *Requiring* $Q = 2/3$ *for neutrinos gives* $Q_{\max} = 0.585$ *(normal)* / 0.500 *(inverted), both* $< 2/3$*; the measured* Δm^2 *exclude all-lepton Koide. (Confidence 0.90.)*

Chapter 14

Gravity and Emergent GR

14.1 Newton's constant

Proposition 14.1 (Dimensional result)**.** *$G = \ell^2 c^3/\hbar$ is a* unit-consistency *result, not a derivation of gravity's dynamics.*

14.2 The phase-gravity chain

oscillation → phase → causal density → metric → curvature → Einstein.

Theorem 14.2 (Leading-order recovery)**.**

$$G_{\mu\nu} = 8\pi G_{\text{eff}}\, T_{\mu\nu} + O(\ell_P^2 R^2), \tag{14.1}$$

with Planck-scale corrections $\sim 10^{-40}$. Newtonian $1/r^2$, lensing, redshift, perihelion precession, and two gravitational-wave polarizations at speed c are reproduced.

Two modifications beyond GR: (1) time-varying dark energy

$$\Lambda(t) = \frac{\alpha}{\sqrt{V(t)}}, \tag{14.2}$$

where $V(t)$ is the 4-volume of the past light cone; and (2) singularity resolution at $r \sim \ell_P$ (maximum curvature $1/\ell_P^2$).

Remark 14.3 (Honesty note)**.** *The chain is* ***logical, not a new dynamical derivation****: "same equations, same predictions; no falsifiable difference" in the weak field. The value is* ontological *(identifying the gravitational potential with the phase field); the physical content beyond GR is confined to $\Lambda(t)$ and singularity resolution.*

14.3 The radial acceleration relation

Theorem 14.4 (Zero-parameter RAR)**.**

$$g_\dagger = \frac{cH_0}{2\pi} \approx 1.05 \times 10^{-10}\ \text{m/s}^2, \tag{14.3}$$

matching the measured a_0. (The factor 2π is a numerical accident, Phase 144.)

Chapter 15

Cosmology

- **Expansion** is an interpretation of FLRW (QG-081); every reinterpretation collapses to FLRW (QG-080–089).
- **Causal-set** Λ: $\Lambda \sim 1/\sqrt{N}$ is a postdiction (Phase 140).
- **Pantheon+SH0ES**: TQM and ΛCDM are indistinguishable at the current signal $w(z) = -1 + 0.015(1+z)^{3/2}$ (DATA-001); a detectability audit (DATA-002) quantified the power required for separation.
- **Clusters**: Coma dynamical mass and ACCEPT X-ray gas fraction are reproduced.
- **CMB**: an accepted partial computational layer; full C_ℓ solver deferred (computational, not a theory gap).

Part IV

Classification, No-Gos, and Predictions

Chapter 16

The Three-Category Taxonomy

Definition 16.1 (Taxonomy)**.**

Category	*Meaning*
DERIVED	*computable from the primitives by theorem*
REAL-UNDERIVED	*real, precise, predictive structure whose origin is not computable*
DRAWN	*a coincidental draw of the abundance law*

The taxonomy classifies underivability*;* ***REAL-UNDERIVED*** *and* ***DRAWN*** *are* not *claimed as derivations.*

16.1 The fourteen classified results

Object	Category	Confidence
$U(1)$	**DERIVED**	0.95
spatial 3	**DERIVED**	0.85
$N \geq 3$	**DERIVED**	0.90
log-normal law	**DERIVED**	theorem
$SU(2)$	**REAL-UNDERIVED** (emergent)	0.70
$SU(3)$ structure	**REAL-UNDERIVED** (emergent)	0.10
Koide $Q = 2/3$ (reality)	**REAL-UNDERIVED** (structured)	0.90
Koide 45° (origin)	**REAL-UNDERIVED** (structured)	0.70
Yukawas / couplings / Ω_{DM}	**DRAWN**	—
$N \leq 3$	**DRAWN**	0.70
color count 3	**DRAWN**	—
internal $N = 3$	**DERIVED$\cap$DRAWN**	0.70
neutrino-Koide	FALSIFIED	0.90

Category conflicts: **0**. Composite objects: **2**. Overall consistency: **0.95**.

Chapter 17

The No-Go Theorems

The no-go theorems are *conditional* — relative to the no-new-primitives constraint and the tested route space — not absolute logical impossibilities (except the computation T-11).

Theorem	Statement
T-08 Koide no-go	no symmetry/attractor/topology/information-geometry selects $Q = 2/$
T-09 Gauge-count no-go (provisional)	no principle fixes n; graph-spectrum/lattice-mode untested
T-10 $N \le 3$ no-go	no stability/anomaly/representation bound gives $N \le 3$
T-11 Neutrino-Koide falsification	$Q_{\max} = 0.585/0.500 < 2/3$ (computation)
T-12 Shared-cascade no-go	one vs three cascades untestable from one universe

Remark 17.1 (Internal-3 disposition)**.** *Gauge count $n = 3$ and multiplicity $N \le 3$ are one node with two formally unlinked faces (N vs n; assumption A-10). It is dispositioned* unresolved-contingent*: the multiplicity face closed (T-10, 0.70); the gauge-count face* provisionally *closed (T-09, 0.10). This is the theory's one open door.*

Chapter 18

Quantitative Predictions

Prediction	Type	Status
RAR $g_\dagger = cH_0/(2\pi) \approx 1.05 \times 10^{-10}$	zero-parameter	matches a_0
$w(z) = -1 + 0.015(1+z)^{3/2}$	cosmological	Euclid $> 3\sigma$ by $\sim$2030
$\Lambda(t) = \alpha/\sqrt{V(t)}$	dark energy	Euclid / Roman
log-normal abundance law	distribution form	testable
$N \geq 3$ (CP lower bound)	a priori	confirmed
neutrino-Koide $Q = 2/3$	falsifiable	**falsified**

The theory is *falsifiable*: it made a concrete prediction (neutrino-Koide) that was falsified, and it carries live quantitative predictions that distinguish it from SM +ΛCDM in the *form* sector, even though the *content* is DRAWN.

Chapter 19

Scope and Limitations

1. The gauge-count face of the internal-3 node is only provisionally closed (T-09, 0.10).
2. Encyclopedia completeness $\approx 81\%$; the CMB chapter is a documented partial computational layer.
3. Content is contingent by design; the theory predicts the distribution *shape*, not the *values*.
4. The shared-cascade question is logically (not empirically) closed.
5. The unified action is a roadmap, not a result.
6. Confidence numbers are uncalibrated ordinal ranks (Phase-149/156 estimates).
7. The phase-gravity chain is ontological in the weak field.

Part V

Verification

Chapter 20

The Executable Verification Program

All claims below are backed by executable xUnit tests (all **PASSING**). The complete inventory is in Appendix A.

20.1 Summary of verified results

Chain link	Test file	Result
$L_Q \to -\nabla^2$	GraphLaplacianContinuumTests	exact spectrum, 4×/doubling
BDG$\to \Box$	BDGOperatorContinuumTests	$O(h^2)$, 4×/halving
L_Q vs BDG signature	QuantumGravityBridgeTests (3)	PSD vs indefinite
no native Δ_g	CurvedSpaceBridgeTests (3)	bridge absent
weighted Laplacian	WeightedLaplacianTests (4)	symmetric/PSD/zero-sum
$L_W \to \Delta_g$	LaplaceBeltramiTests (3)	S^1 example
curved Schrödinger	CurvedSchrodingerTests (3)	unitary
Einstein recovery	EinsteinRecoveryTests (3)	PARTIAL
Einstein chain	EinsteinTensorTests (4)	standard 2D/3D
Einstein integration	EinsteinTensorIntegrationTests (4)	builder works
metric generation	MetricGenerationTests (4)	distance PRESENT
metric emergence	MetricEmergenceTests (4)	factor native
conformal structure	ConformalStructureTests (3)	class imported
metric origin	MetricOriginTests (3)	closure

20.2 The decisive findings

1. The *Schrödinger* side of the continuum limit is **controlled and tested**: $L_Q \to -\nabla^2 \to$ Schrödinger (and the curved L_W, and the Laplace–Beltrami approximation) are verified.

2. The *Einstein* side is **partially** verified: the standard $g \to G_{\mu\nu}$ chain works, but TQM has no native metric (Malament import), and the BDG action is imported.

3. The conformal-class "gap" is an *imported theorem* (Malament), not a theory gap.

Chapter 21

The Hostile-Review Audit Trail

A hostile referee (Round 2) raised twelve FATAL issues. Each was re-evaluated against the executable tests. The classifications are:

Round-2 FATAL issue	Classification
Primitives undefined as mathematical objects	PARTIALLY RESOLVED
Schrödinger derivation circular/incomplete	RESOLVED
Gauge derivations = elementary topology	PARTIALLY RESOLVED
Complexity argument not a derivation	PARTIALLY RESOLVED
Structure/content split = immunization	PARTIALLY RESOLVED
Composite objects admitted	PARTIALLY RESOLVED
Internal-3 unresolved while "complete"	RESOLVED
T-09 at 0.10 cannot close	RESOLVED
"Structurally complete" overstatement	RESOLVED
Gravity→Einstein is ontological	RESOLVED
$\mathrm{Aut}(S^1) = U(1)$ as EM derivation	RESOLVED
No controlled continuum limit	PARTIALLY RESOLVED

Tally: RESOLVED = 6 · PARTIALLY RESOLVED = 6 · OPEN = 0.

21.1 What changed

The framing overstatements ("closed" vs. "dispositioned", "no-go" vs. "conditional no-go", "derivation" vs. "ontological reinterpretation") were corrected in the revised paper. The *substantive* objection — "no controlled continuum limit" — was met on the Schrödinger side (now tested: $L_Q \to -\nabla^2 \to$ Schrödinger, and curved L_W, Laplace–Beltrami, unitary), and *partially* on the Einstein side (the standard chain works, but the metric and BDG action are imported).

21.2 The decisive residual

The single genuinely-open item is the **native metric → operator coupling** (the “G4” gap): TQM imports $g_{\mu\nu}$ (Malament) rather than generating it from Q-events. The metric *origin* is closed (Chapter 10); the metric *dynamics* remain imported.

Verdict: READY_FOR_WHITEPAPER — NOT_READY_FOR_JOURNAL (see Chapter 22).

Chapter 22

Conclusion

THE Q-MODEL is *structurally complete* in the precise sense that every in-scope question is **dispositioned** — derived, underivable, underdetermined, contingent, or accepted as a computational layer.

The single residual of genuine scientific tension is the **internal-3 node** — why the internal multiplicity/count saturates at 3 — dispositioned unresolved-contingent with a provisional gauge-count no-go (T-09, 0.10). This is the theory's one open door.

Structure is derivable. Content is realized.

Publication status: READY_FOR_WHITEPAPER — NOT_READY_FOR_JOURNAL. The Schrödinger dynamical core is executable and tested; the Einstein recovery remains logical, not mathematical (imported metric + imported BDG action, G dimensional, no unique discriminating prediction).

Appendix A

Complete Test Inventory

Test file	Tests	Verified quantity	Status
GraphLaplacianContinuumTests	1	$\lambda_k = (1/\Delta x^2)[2 - 2\cos(\pi k/(N{+}1))] \to (\pi k)^2$	PASS
BDGOperatorContinuumTests	1	BDG$\to \Box$, $O(h^2)$	PASS
QuantumGravityBridgeTests	3	L_Q PSD; BDG indefinite; incompatible	PASS
CurvedSpaceBridgeTests	3	no Δ_g; $\Delta_g \to \nabla^2$; absent	PASS
MetricOperatorTests	4	candidate constructions	PASS
WeightedLaplacianTests	4	symmetric; zero-sum; PSD; unweighted limit	PASS
LaplaceBeltramiTests	3	flat limit; weights; S^1 ($2k{-}1$ degeneracy)	PASS
CurvedSchrodingerTests	3	unitary; flat limit; curved operator	PASS
EinsteinRecoveryTests	3	PARTIAL (no full $G_{\mu\nu}$)	PASS
EinsteinTensorTests	4	flat/sph Γ, K, R, G	PASS
EinsteinTensorIntegrationTests	4	builder; 3-sphere $G = -g$	PASS
MetricGenerationTests	4	distance PRESENT; candidate PARTIAL	PASS
MetricEmergenceTests	4	factor $f = \rho^{2/d}$; $R = -0.6145$	PASS
ConformalStructureTests	3	axioms; null invariant; Malament	PASS
MetricOriginTests	3	class unique; factor free; imported	PASS

Appendix B

Confidence and Classification Tables

The confidence numbers are uncalibrated ordinal ranks (Phase-149/156 estimates), not posterior probabilities. They rank, they do not measure.

B.1 Confidence table

Result	Confidence	Note
program consistency	0.95	zero category conflicts
$U(1)$	0.95	theorem
$N \geq 3$	0.90	CP lower bound
neutrino-Koide	0.90	falsification (computation)
spatial 3	0.85	window intersection
$SU(2)$	0.70	emergent
Koide 45°	0.70	structured
$N \leq 3$	0.70	no-go
$SU(3)$ structure	0.10	provisional no-go

B.2 Final classification

Question	Classification
Is the metric origin closed?	YES (imported Malament, proven)
Is the Schrödinger continuum limit controlled?	YES (tested)
Is the Einstein recovery a derivation?	NO (logical, imported)
Is TQM journal-ready?	NO
Is TQM white-paper-ready?	YES

Appendix C

Worked Derivation: The Einstein Chain on the 2-Sphere

This appendix exhibits the standard differential-geometry chain explicitly, so the reader can follow every step without opening the repository code.

C.1 Setup

The unit 2-sphere has metric (coordinates $x^1 = \theta$, $x^2 = \phi$)

$$g = \mathrm{diag}(g_{\theta\theta},\ g_{\phi\phi}) = \mathrm{diag}\Big(1,\ \sin^2\theta\Big), \qquad g^{\theta\theta} = 1, \quad g^{\phi\phi} = \frac{1}{\sin^2\theta}. \tag{C.1}$$

C.2 Christoffel symbols

$$\Gamma^{\lambda}_{\mu\nu} = \tfrac{1}{2} g^{\lambda\sigma} \left(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu} \right). \tag{C.2}$$

The only non-vanishing partial derivative is $\partial_\theta g_{\phi\phi} = 2\sin\theta\cos\theta$. Therefore

$$\Gamma^{\theta}_{\phi\phi} = -\tfrac{1}{2}\, g^{\theta\theta}\, \partial_\theta g_{\phi\phi} = -\sin\theta\cos\theta, \tag{C.3}$$

$$\Gamma^{\phi}_{\theta\phi} = \Gamma^{\phi}_{\phi\theta} = \tfrac{1}{2}\, g^{\phi\phi}\, \partial_\theta g_{\phi\phi} = \frac{\cos\theta}{\sin\theta} = \cot\theta. \tag{C.4}$$

At $\theta = \pi/4$: $\Gamma^{\theta}_{\phi\phi} = -1/2$, $\Gamma^{\phi}_{\theta\phi} = 1$.

C.3 Riemann curvature

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu \Gamma^{\rho}_{\nu\sigma} - \partial_\nu \Gamma^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma}. \tag{C.5}$$

The single independent component is

$$R^{\theta}{}_{\phi\theta\phi} = \partial_\theta \Gamma^{\theta}_{\phi\phi} - \Gamma^{\theta}_{\phi\phi}\Gamma^{\phi}_{\theta\phi} \tag{C.6}$$

$$= -\Big(\cos^2\theta - \sin^2\theta\Big) + \sin\theta\cos\theta \cdot \cot\theta \tag{C.7}$$

$$= \sin^2\theta. \tag{C.8}$$

At $\theta = \pi/4$: $R^{\theta}{}_{\phi\theta\phi} = 1/2$.

C.4 Gauss curvature, Ricci, and Einstein

Gauss curvature:

$$K = \frac{R^{\theta}{}_{\phi\theta\phi}}{g_{\phi\phi}} = \frac{\sin^2\theta}{\sin^2\theta} = 1. \tag{C.9}$$

Ricci tensor (in 2D, $R_{\mu\nu} = K g_{\mu\nu}$):

$$R_{\theta\theta} = K g_{\theta\theta} = 1, \qquad R_{\phi\phi} = K g_{\phi\phi} = \sin^2\theta = \tfrac{1}{2} \text{ at } \theta = \pi/4. \tag{C.10}$$

Ricci scalar:

$$R = g^{\mu\nu} R_{\mu\nu} = 2K = 2. \tag{C.11}$$

Einstein tensor (vanishes identically in 2D):

$$G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2} R g_{\mu\nu} = 0. \tag{C.12}$$

All of these are reproduced numerically by `EinsteinTensorBuilder` to within 10^{-3}. A non-trivial $G_{\mu\nu}$ requires dimension ≥ 3; the unit 3-sphere gives $G_{\mu\nu} = -g_{\mu\nu}$ (Appendix A).

Appendix D

Research Programs

The derivation program is organized into research programs, each closed by executable experiments:

Program	Scope	Experiments
DATA	Cosmology & RAR	10 (DATA-001→010)
QM	Quantum foundations	5 (QM-001→005)
QG	Quantum gravity	31 (QG-001→031)
XC	Continuum-limit bridge	(audits + tests)

Key results of the full program:

- G is not fundamental: $G = \ell^2 c^3/\hbar$ (gravity emerges from the triple).
- $(\ell, \tau, \hbar)$ is the irreducible triple — one process, three aspects.
- Oscillation is a logical inevitability (cannot be removed without destroying Q).
- Gravity is a phase-gradient phenomenon; attraction dominates, repulsion is dark energy.
- Gravity manipulation is not possible (QG-023→031).
- RAR derivation $g_\dagger = cH_0/(2\pi)$ with zero free parameters.
- Parameter compression SM+GR (∼26) → TQM (2+3), ∼5–8× reduction.
- Novel predictions: $w(z) = -1 + 0.015(1+z)^{3/2}$; evolving $g_\dagger(z)$; Euclid sensitivity.

Appendix E

Glossary

Term	Meaning
Q	principle of individuation (ontology)
Random Actualization	ontological chance (assumption A-03)
$(\ell, \tau, \hbar)$	spacetime scale, clock, action (units)
M^2	nonlinearity parameter (single continuous)
L_Q	graph Laplacian $D - A$
L_W	weighted Laplacian $D_K - K$
BDG	Benincasa–Dowker–Glaser d'Alembertian
Δ_g	Laplace–Beltrami operator
$G_{\mu\nu}$	Einstein tensor $R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$
DERIVED	computable from primitives by theorem
REAL-UNDERIVED	real structure, underivable origin
DRAWN	coincidental draw of the abundance law
T-08. . . T-12	conditional no-go theorems
A-03, A-06, A-07, A-10	assumptions (random actualization; Z_2 lift; gauge dynamics; N/n split)

Appendix F

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